

Dormify — Software Development Plan

Dormitory Management System

Field	Details
Version	1.1 — PA3
Date	23 July 2026
Development process	Scrum
Delivery structure	Five sprints, with Sprint 1–5 corresponding to PA1–PA5
Repositories	Dormitory Management Frontend and Backend
Compared version	SDP v1.0 / PA2 baseline

Delivery Snapshot

Review item	Summary
Current assignment	PA3
Development model	Scrum with five PA-aligned sprints
PA3 focus	UI improvement, AI research, missing Admin and Student workflows, and one end-to-end functional group
Selected functional group	Maintenance Management (FR26 - FR28 , ST17 - ST18 , MT01 - MT08)
Formal build target	Build 3 on 25 July 2026
Operational evidence	Trello board, Git logs, Weekly Report, AI Usage Report, Spec Kit artifacts, Markdown files, and PDF exports

Revision History

Performed by: Trần Huỳnh Mạnh Đạt | Reviewed by: Đào Duy Anh | Edited by: Trần Huỳnh Mạnh Đạt

Date	Version	Description	Author
08 July 2026	1.0	Initial Software Development Plan.	Trần Huỳnh Mạnh Đạt
23 July 2026	1.1	Improve Software Development Plan based on TA's comments.	Trần Huỳnh Mạnh Đạt

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1. Introduction

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This Software Development Plan defines how the Dormify team will plan, implement, review, test, and demonstrate the Dormitory Management System by applying Scrum. Development is organized into five sprints, and each sprint corresponds to one course Project Assignment. The plan is an updated version for PA3 after teaching-assistant feedback and changes in the Trello board.

The previous version was an initial PA2 planning baseline. This PA3 revision incorporates the teaching-assistant feedback as follows:

Feedback area	PA3 response
Sprint planning detail	Adds detailed tasks for every sprint instead of only the current sprint.
Task traceability	Adds owner, reviewer, estimate, dependency, due date, acceptance criterion, Jira/Trello ID, and evidence fields where applicable.
Scope alignment	Aligns the project scope with the revised Use-Case Model.
PA3 deliverables	Adds <code>Changes.md</code> , use-case specifications, prototypes, Spec Kit artifacts, source code, and the narrated demo video.
AI reporting	Expands the AI Usage Report and its audit trail requirements.
Document quality	Improves Markdown and PDF consistency.

1.1 Purpose

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The purpose of this plan is to:

- Establish the Product Goal, sprint structure, team responsibilities, and delivery rules.
- Define the Product Backlog, Sprint Backlog, Increment, Definition of Ready, and Definition of Done.
- Provide all detailed information of all Sprints, especially Sprint 3 tasks with one owner, a different reviewer, and a due date.
- Provide planned work for later sprints.
- Define five formal integration builds and the testing purpose of each build.
- Keep the written schedule consistent with Trello.
- Identify project risks and mitigation strategies.

- Ensure that coding, documentation, report writing, testing, and self-training are treated as valid project tasks.

1.2 References

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- PA1-2026 Project Proposal and App Survey.
- PA2-2026 Project Assignment.
- PA3-2026 Project Assignment.
- NHỮNG TÍNH NĂNG CHÍNH CỦA HỆ THỐNG QUẢN LÝ KTX.
- Backend repository: [Dormitory Management Backend](#).
- Frontend repository: [Dormitory Management Frontend](#).
- Trello tracking log.
- Weekly Reports, AI Usage Report, Spec Kit artifacts, and training evidence.

2. Project Overview

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2.1 Product Goal

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The Product Goal is to deliver a secure, role-based web application that centralizes primary dormitory workflows for students and authorized staff. By the final sprint, the team aims to demonstrate an integrated and testable product covering authentication, rooms and bookings, contracts, invoices, maintenance, absences, violations, notifications, room transfers, and selected dashboard capabilities.

2.2 Project Objectives

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1. Deliver usable Product Increments at the end of every sprint.
2. Maintain one ordered Product Backlog in Trello.
3. Assign every task to one owner and one different reviewer.
4. Integrate frontend and backend work at least twice per sprint.
5. Produce five formal builds for testing and review.
6. Pass all critical acceptance scenarios before the PA5 demonstration.
7. Keep all submitted documents in English and Markdown, with Mermaid diagrams where required.
8. Clearly distinguish implemented, partial, planned, and future capabilities.

2.3 Technical Baseline

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Layer	Technology or tool	Responsibility
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Frontend	Next.js and TypeScript	Responsive role-based screens and API integration
Backend	NestJS and TypeScript	REST APIs, validation, authorization, and business logic
Database	MongoDB	Persistent user, room, contract, invoice, request, and audit data
Real-time communication	Socket.IO	Notifications and status updates where applicable
Authentication	JWT and Google OAuth	Login, authorization, and external authentication
File storage	Cloudinary or approved storage service	Prototype and maintenance image evidence
Planning	Scrum and Trello	Backlog ordering, sprint planning, and task tracking
Specification	Spec Kit and Markdown	Specification, planning, and task artifacts

2.4 Assumptions and Constraints

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ID	Assumption or constraint	Response
A-01	Team members remain available during the academic schedule.	Forecast capacity at Sprint Planning and re-plan incomplete work.
A-02	Dormitory rules can be clarified by the project representative or lecturer.	Do not commit unclear items without acceptance criteria.
A-03	Trello contains the latest task owner, reviewer, date, estimate, and status.	Treat Trello as the operational source of truth.
A-04	Frontend and backend are maintained in separate repositories.	Link related frontend and backend changes to one task or story.
A-05	Real payment credentials may not be available.	Use a mock or sandbox payment flow and label real gateways as planned.
A-06	The system handles sensitive student information.	Use synthetic data, role guards, secret management, and negative authorization tests.
C-01	PA deadlines are fixed.	Postpone lower-priority features before reducing critical quality work.
C-02	Some features are specified but not implemented in the current increment.	Mark each capability as implemented, partial, planned, or removed.

3. Product Scope and Feature Traceability

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

3.1 Actors and Responsibility Boundaries

Actor	Responsibility
Applicant / Guest	Submits preliminary residence information and starts the accommodation process.
Student	Manages personal information, room applications, contracts, invoices, residence declarations, repair requests, feedback, and conduct information.
System Admin	Manages accounts, roles, permissions, audit logs, and system administration.
Dormitory Manager	Manages rooms, students, contracts, checkout, finance, residence, maintenance, feedback, and conduct.
Maintenance Staff	Views assigned maintenance work and updates repair status, results, and evidence.
Google OAuth Provider	Supports Google authentication.
Email / SMS Service	Sends contact notifications and password-reset codes.
Payment Gateway	Processes supported online payments.
Scheduled Trigger	Starts time-based jobs such as overdue processing and reminders.

The previously proposed **Floor Manager** role has been removed. Its necessary responsibilities are assigned to the Dormitory Manager.

3.2 Functional Scope Matrix

Functional group	Requirements and features	PA3 status	Related evidence
Authentication and profile	FR01-FR03, ST01-ST06	Specified; implementation status must be verified	Auth module, Use-Case Model
System administration	FR04-FR07	Specified; implementation status must be verified	Admin screens and permissions
Room and student management	FR09-FR13, ST07-ST10	Core project scope	Room module and room application tests
Contract management	FR14-FR17, ST11-ST12	Planned or partial	Contract specifications and PDF evidence
Checkout and deposit refund	FR18, FR20-FR21	Planned or partial	Checkout workflow
Finance and invoices	FR22-FR24, ST13-ST16	Planned or partial	Invoice APIs, payment mock, reports
Residence management	FR25, ST21-ST24	Specified; implementation status must be verified	Residence declarations

Maintenance management	FR26-FR28, MT01-MT08, ST17-ST18	Selected PA3 end-to-end functional group	Maintenance specification and implementation
Conduct and evaluation	FR29-FR30, ST27-ST28	Planned or partial	Conduct records and rules
Feedback and notifications	ST19-ST20, ST25-ST26	Specified; implementation status must be verified	Feedback and notification workflows

3.3 Scope Decisions

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

- **FR08 - Back up and restore data** is removed from the current user-facing project scope.
- **FR19 - Inspect assets** is removed as a standalone use case. Damaged or missing items may be entered manually during checkout when compensation is calculated.
- **FR24 - Generate revenue reports** remains in scope as a planned management capability.
- The Floor Manager actor is removed. Meter, residence, room, conduct, feedback, and repair-monitoring responsibilities are assigned to the Dormitory Manager.
- Real payment providers remain planned unless the team has a working sandbox integration.
- A capability is described as implemented only when code, review, and validation evidence exist.

Planned or Partial Capabilities

- Automatic room allocation based on student preferences.
- Contract renewal, termination, check-in/check-out, and verified PDF export.
- Real payment-provider integration.
- Meter readings, meter images, and consumption history.
- Maintenance assignment, before-and-after images, staff history, and a maintenance dashboard.
- Visitor registration, full temporary-residence procedures, feedback, complaints and activity logs.
- Administrative backup/restore has been removed from the committed project scope.

Outside the Committed PA3 Scope

- Native mobile applications.
- Offline synchronization.
- Physical access-control hardware.
- Broad integration with unrelated university systems.
- AI maintenance classification and a rules chatbot unless they are approved, estimated, and added to the Product Backlog without displacing critical scope.

4. Scrum Organization

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

4.1 Scrum Roles

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Scrum role	Assigned member	Responsibilities
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Scrum Master	Đào Duy Anh	Facilitates Scrum events, removes impediments, and maintains planning discipline.
Backend / Database	Trần Huỳnh Mạnh Đạt, Đào Duy Anh and Trần Hoàng Quốc Khánh	APIs, authorization, business rules, database operations, and backend tests.
Frontend / UI/UX	Hồ Phúc Kiên and Đào Duy Anh	Responsive screens, validation, API integration, and usability.
Quality and Testing	Trần Huỳnh Mạnh Đạt	Test design, acceptance evidence, regression testing, and defect reporting.
Documentation and Analysis	Trần Huỳnh Mạnh Đạt	Vision, SDP, traceability, reports, and submission artifacts.
Research and Training	Trần Hoàng Quốc Khánh	Training evidence, approved research, and AI report support.

Team Membership Update

Member	Updated status	Reason	Planning impact
Tô Trần Hoàng Triệu	Removed from the active project group in the PA3 revision.	The member did not provide meaningful contribution, repeatedly missed agreed deadlines, and submitted work that required significant correction and rework by other members.	Future responsibilities and review duties are reassigned to active members; prior evidence involving this member must be checked before submission.

4.2 Scrum Artifacts and Events

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Artifact or event	Required content or result	Evidence
Product Backlog	Epics, user stories, bugs, research, documentation, training, and technical tasks ordered by value and risk.	Trello board export
Sprint Backlog	Selected tasks with owner, reviewer, estimate, dependency, due date, acceptance criteria, and status.	Sprint board screenshot
Product Increment	Integrated and reviewed work that satisfies the Definition of Done.	Build record and demonstration
Sprint Planning	Sprint Goal, selected backlog items, capacity, dependencies, and build target.	Weekly Report
Scrum meeting	Progress, next work, impediments, integration needs, and risks.	Meeting notes
Sprint Review	Demonstrated increment, feedback, acceptance decisions, and backlog updates.	Review record and screenshots
Sprint Retrospective	Improvement actions for the next sprint.	Retrospective notes

4.3 Definition of Ready

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

A backlog item is ready when its value and actor are clear, acceptance criteria are testable, dependencies are identified, one owner and one different reviewer are assigned, and the team can estimate and complete it within a sprint.

4.4 Definition of Done

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

A task is done when its acceptance criteria are satisfied, the result is committed, a different team member has reviewed it, relevant tests pass, security effects are considered, documentation is updated, Trello evidence is linked, and no unresolved critical defect prevents use.

5. Five-Sprint Product Plan

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Sprint	Sample dates	Assignment	Sprint Goal	Main increment
Sprint 1	23 May – 06 June 2026	PA1	Establish the product concept and project workspace.	Approved proposal, survey, repositories, and initial prototype.
Sprint 2	06 June – 11 July 2026	PA2	Establish authentication, authorization, profile, room-management foundation, and planning baselines.	Authentication/core workflows, Vision, SDP, Spec Kit setup, reports, and baseline integration.
Sprint 3	11 – 25 July 2026	PA3	Improve the UI, research AI capabilities, complete missing Admin and Student features, and implement one end-to-end functional group.	Improved UI, AI research/POC, Admin and Student workflows, Use-Case Model, specifications, Maintenance Management increment, and Build 3.
Sprint 4	25 July – 08 August 2026	PA4	Integrate finance, residence, conduct, notifications, and operational workflows.	Operational Beta and Build 4.
Sprint 5	08 – 22 August 2026	PA5	Harden, deploy, and demonstrate the accepted product.	Final Product Increment, UAT, release evidence, and Build 5.

5.1 Sprint Task Standard

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Every sprint backlog uses a consistent task schema so reviewers can scan ownership, review responsibility, traceability, estimates, dependencies, acceptance criteria, and delivery status.

Field	Display rule
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Task ID	Use the sprint-prefix key from Trello or the SDP backlog.
Requirement / UC	Link the task to the relevant functional requirement, story, use case, or assignment item when applicable.
Detailed task	State the work as a concrete action with an observable outcome.
Owner	Assign exactly one accountable owner.
Reviewer	Assign one reviewer who is different from the owner.
Estimate	Record the planned effort in hours.
Dependency	Identify prerequisite work or evidence when the task depends on another item.
Acceptance criteria	Define how the team will know the task is complete.
Status	Use a consistent state such as Completed, In Progress, or Planned.

Status	Meaning
Completed	Acceptance criteria are satisfied and review evidence is available.
In Progress	Work has started and still needs completion, review, or final evidence.
Planned	Work is forecast for a later sprint or has not started yet.

5.2 Sprint 1 Backlog – Project Foundation

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Task ID	Detailed task	Owner	Reviewer	Estimate	Acceptance criteria	Status
S1-PRO-01	Define the Dormify problem, target users, and business goals.	Đạt	Duy Anh	4h	Proposal contains approved problem and target-user statements.	Completed
S1-PRO-02	Survey existing dormitory-management applications and alternatives.	Kiên	Duy Anh	5h	Survey evidence and comparison table are included.	Completed
S1-PRO-03	Create frontend and backend repositories with a documented baseline.	Duy Anh	Đạt	3h	Both repositories can be cloned and started.	Completed
S1-PRO-04	Create the initial feature list and actor list.	Đạt	Duy Anh	4h	Initial features are recorded and traceable to the proposal.	Completed
S1-PRO-05	Create the initial prototype and project presentation.	Khánh	Triệu	6h	Prototype demonstrates the selected product direction.	Completed

5.3 Sprint 2 Backlog – PA2 Planning Baseline

Task ID	Requirement / UC	Detailed task	Owner	Reviewer	Estimate	Dependency	Acceptance criteria	Status
S2-DOC-01	PA2	Audit PA2 requirements against the Vision Document and SDP.	Đạt	Duy Anh	5h	PA2 rubric	Compliance checklist is complete.	Completed
S2-AUTH-01	FR02-FR03	Define the user, credential, role, permission, and session data model.	Duy Anh	Đạt	5h	Repository baseline	Required fields and role boundaries are documented.	Completed
S2-AUTH-02	FR01	Implement preliminary residence-profile submission for an Applicant / Guest.	Duy Anh	Đạt	6h	S2-AUTH-01	Valid applicant information is stored and an acknowledgment is returned.	Completed
S2-AUTH-03	FR02	Implement email/CCCD login, logout, JWT validation, and Google OAuth flow.	Duy Anh	Đạt	10h	S2-AUTH-01	Valid users can log in and invalid credentials are rejected.	Completed
S2-AUTH-04	FR02	Implement forgotten-password and OTP verification flow.	Đạt	Khánh	6h	S2-AUTH-03	OTP expiry, invalid OTP, and successful password reset are tested.	Completed
S2-PRO-01	FR03	Implement view profile, update contact information, and change password.	Duy Anh	Đạt	6h	S2-AUTH-03	Users can update allowed fields and unauthorized identity changes are blocked.	Completed

S2-ADM-01	FR05	Create the initial role and permission matrix for Admin, Student, Dormitory Manager, and Maintenance Staff.	Đạt	Duy Anh	4h	S2-AUTH-01	Each protected feature has an identified permitted role.	Completed
S2-ROOM-01	FR09	Define building, floor, room, bed, capacity, and occupancy data structures.	Duy Anh	Triệu	6h	S2-AUTH-01	Room capacity and bed availability can be represented consistently.	Completed
S2-ROOM-02	ST07-ST08	Implement room search, filtering, room details, and available-bed display.	Kiên	Duy Anh	8h	S2-ROOM-01	Student can filter by room type, building, price, and available beds.	Completed
S2-NOT-01	ST25-ST26	Define the notification model and basic user-notification delivery path.	Duy Anh	Triệu	5h	S2-AUTH-03	Notifications are associated with the correct authorized user.	Completed
S2-DOC-02	PA2	Write the Vision Document with functional and non-functional requirements.	Đạt	Duy Anh	10h	S2-DOC-01	Required sections and quality targets are present.	Completed
S2-DOC-03	PA2	Write the SDP around Scrum roles, artifacts, events, and five sprints.	Đạt	Duy Anh	8h	S2-DOC-01	SDP contains a consistent schedule and evidence rules.	Completed

S2-SPEC-01	PA2	Initialize Spec Kit and store the generated artifacts in the approved project path.	Duy Anh	Khánh	4h	Repository baseline	Spec, plan, and task files are committed.	Completed
S2-TEST-01	FR01-FR03	Test registration, login, logout, OTP, profile authorization, and core room queries.	Đạt	Triệu	8h	S2-AUTH-04, S2-PRO-01, S2-ROOM-02	Test results and known limitations are recorded.	Completed
S2-REPORT-01	PA2	Prepare the AI Usage Report and Weekly Report.	Duy Anh	Khánh	6h	Sprint evidence	Reports contain required logs and screenshots.	Completed
S2-EVID-01	PA2	Export Git log, Trello screenshots, and the PA2 package.	Đạt	Duy Anh	3h	Final documents	Evidence is included in the PA2 folder.	Completed

5.4 Sprint 3 Backlog – PA3 and Maintenance Management

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Selected PA3 functional group: Maintenance Management (FR26-FR28 , ST17-ST18 , and MT01-MT08).

Workstream	Task IDs	Review focus
Requirements and specification	S3-REQ, S3-UC, S3-SPEC	Maintenance actors, lifecycle, use cases, and complete specifications.
UI and shared experience	S3-UI, S3-STU, S3-FE	Consistent screens, shared components, student workflows, and staff workspaces.
Admin capabilities	S3-ADM	Account, permission, audit log, and dashboard coverage.
Maintenance implementation	S3-DB, S3-BE, S3-INT, S3-TEST	Data model, APIs, integration, image evidence, authorization, and test results.
AI research	S3-AI	Research, dataset definition, RAG architecture, risk review, and POC evidence.
Submission evidence	S3-PA3	Changes .md , reports, screenshots, Git/Trello evidence, Markdown, and PDF outputs.

Task ID	Requirement / UC	Detailed task	Owner	Reviewer	Estimate	Dependency	Acceptance criteria	Status
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S3-REQ-01	FR26-FR28	Refine maintenance actors, request lifecycle, priorities, and statuses.	Đạt	Duy Anh	4h	Vision feedback	Approved requirements identify Student, Dormitory Manager, and Maintenance Staff flows.	Completed
S3-UI-01	All roles	Audit the current UI for inconsistent spacing, typography, colors, navigation, and terminology.	Kiên	Đạt	5h	Existing screens	UI issue list is prioritized by user impact.	Completed
S3-UI-02	All roles	Create a shared design system for buttons, forms, tables, cards, status badges, and modal dialogs.	Kiên	Duy Anh	8h	S3-UI-01	Shared components are reused in at least the Student and Admin screens.	Completed
S3-UI-03	All roles	Add consistent loading, empty, validation-error, success, and unauthorized states.	Kiên	Đạt	6h	S3-UI-02	Every selected workflow has documented state handling.	Completed
S3-UI-04	Admin / Student	Improve dashboard navigation, filters, data tables, and accessibility labels.	Kiên	Duy Anh	8h	S3-UI-03	Main screens pass the team's visual and accessibility checklist.	Completed
S3-AI-01	AI-01	Research NLP approaches for maintenance-category and priority classification.	Khánh	Duy Anh	6h	S3-REQ-01	Research compares feasible models, data needs, latency, cost, and limitations.	Planned

S3-AI-02	AI-01	Define a labeled sample dataset and taxonomy for Plumbing, Electrical, Carpentry, Normal, and Urgent tickets.	Đạt	Khánh	5h	S3-AI-01	Dataset format and labeling rules are documented with example records.	Completed
S3-AI-03	AI-02	Research RAG architecture, embedding storage, chunking, retrieval, citations, and fallback behavior.	Khánh	Duy Anh	8h	Approved rulebook sources	Architecture decision and data-flow diagram are documented.	In Progress
S3-AI-04	AI-01, AI-02	Review privacy, hallucination, prompt-injection, confidence, human-override, and escalation risks.	Khánh	Duy Anh	5h	S3-AI-01, S3-AI-03	AI risk register and mitigation rules are approved.	Completed
S3-AI-05	AI-01, AI-02	Create a small proof-of-concept or mock evaluation for both proposed AI features.	Duy Anh	Khánh	8h	S3-AI-02, S3-AI-03	POC output, test queries, limitations, and recommendation are recorded.	Planned
S3-ADM-01	FR04	Implement Admin account list, search, view, activate, deactivate, lock, and unlock actions.	Duy Anh	Đạt	8h	S2-AUTH-03	Admin can manage accounts according to the permission matrix.	Completed
S3-ADM-02	FR05	Implement role and permission assignment for System Admin and Dormitory Manager.	Duy Anh	Triệu	8h	S2-ADM-01	Unauthorized users cannot change roles or permissions.	Completed

S3-ADM-03	FR06	Implement audit-log recording, filtering, and detail view.	Đạt	Duy Anh	8h	S3-ADM-01	Important account and permission actions produce searchable audit entries.	Completed
S3-ADM-04	FR07	Improve the administration dashboard with account, role, warning, and activity summaries.	Kiên	Đạt	6h	S3-ADM-01, S3-UI-05	Dashboard values match approved backend data.	Completed
S3-STU-01	ST01-ST06	Complete student profile, contact information, current residence, roommate, and residence-history screens.	Kiên	Duy Anh	8h	S2-PRO-01	Student sees only their authorized personal and residence information.	Completed
S3-STU-02	ST07-ST10	Complete room details, application submission, application status, and room-transfer status screens.	Kiên	Đạt	10h	S2-ROOM-02	Student can submit and track an approved room workflow.	Completed
S3-STU-03	ST11-ST12	Complete contract viewing and contract PDF download screens.	Kiên	Triệu	6h	Contract API	Student can view only their own contract and download the approved file.	Completed
S3-STU-04	ST13-ST16	Complete invoice viewing, payment, payment history, and invoice PDF screens.	Kiên	Duy Anh	8h	Invoice API	Student can see invoice status and payment history correctly.	Completed

S3-STU-05	ST17-ST18	Complete student maintenance-request submission, attachment, and progress tracking.	Kiên	Đạt	8h	S3-REQ-01	Student can submit a repair request and track its status.	Completed
S3-STU-06	ST19-ST24	Complete feedback, suggestion, overnight absence, temporary residence, temporary absence, and visitor screens.	Kiên	Khánh	10h	Residence and feedback requirements	Each submission has validation, status, and confirmation behavior.	Completed
S3-STU-07	ST25-ST26	Complete notification list, read state, and message-center screens.	Kiên	Triệu	6h	S2-NOT-01	Student receives and reads authorized notifications.	Completed
S3-STU-08	ST27-ST28	Complete evaluation-history and violation-history screens.	Kiên	Đạt	5h	Conduct API	Student can view only their own records.	Completed
S3-UC-01	UC-MNT-01 to UC-MNT-10	Create Mermaid maintenance use-case diagram with actors and relationships.	Đạt	Duy Anh	4h	S3-REQ-01	Diagram covers submit, assign, track, update, evidence, history, and dashboard flows.	In Progress
S3-SPEC-01	UC-MNT-01 to UC-MNT-10	Write maintenance basic flows, alternatives, preconditions, postconditions, and special requirements.	Đạt	Kiên	10h	S3-UC-01	Every maintenance use case has a complete specification.	Completed

S3-DB-01	FR26-FR28	Define maintenance request, assignment, status, result, rating, and image fields.	Duy Anh	Đạt	5h	S3-REQ-01	Schema validates required fields and status transitions.	Completed
S3-BE-01	FR26-FR28	Implement the API for submitting and viewing repair requests.	Duy Anh	Triệu	8h	S3-DB-01	Authorized students can create and track their own requests.	Completed
S3-BE-02	MT01-MT04	Implement manager assignment and maintenance-staff status updates.	Duy Anh	Đạt	8h	S3-BE-01	Only authorized roles can assign or update repair work.	Completed
S3-BE-03	MT05-MT08	Implement repair-result, image-evidence, history, and dashboard endpoints.	Duy Anh	Triệu	8h	S3-BE-02	Results and before/after evidence are persisted and retrievable.	Completed
S3-FE-01	MT01-MT08	Implement manager and maintenance-staff workspaces using the shared UI components.	Kiên	Đạt	10h	S3-UI-02, S3-BE-02	Staff can view assigned jobs and update repair progress.	Completed
S3-INT-01	FR26-FR28	Integrate maintenance frontend, backend, database, and image storage.	Triệu	Duy Anh	6h	S3-BE-03, S3-FE-01	End-to-end maintenance workflow persists data after refresh.	Completed
S3-TEST-01	FR26-FR28	Test authorization, validation, status transitions, image upload, and error handling.	Đạt	Triệu	8h	S3-INT-01	All acceptance scenarios pass and defects have reproduction steps.	Completed

S3-PA3-01	PA3	Complete Changes .md , AI Usage Report, Weekly Report, and Git/Trello evidence.	Đạt	Duy Anh	6h	Sprint evidence	All PA3 documents are included in Markdown and PDF form.	Completed
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5.5 Sprint 4 Backlog – Operational Workflows

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Task ID	Requirement group	Detailed task	Owner	Reviewer	Estimate	Acceptance criteria	Status
S4-FIN-01	FR22-FR24, ST13-ST16	Implement invoice generation, invoice viewing, payment mock, and payment history.	Duy Anh	Đạt	10h	Student can view invoices and complete the approved payment flow.	Planned
S4-FIN-02	FR23-FR24	Implement debt summary, overdue processing, reminders, and revenue report data.	Duy Anh	Đạt	8h	Overdue invoices and reminders are generated according to the documented rule.	Planned
S4-RES-01	FR25, ST21-ST24	Implement overnight absence, temporary residence, temporary absence, and visitor workflows.	Kiên	Đạt	8h	Student submissions are stored and visible to the Dormitory Manager.	Planned
S4-COND-01	FR29-FR30, ST27-ST28	Implement violation records, conduct-score rules, and evaluation history.	Đạt	Duy Anh	8h	Rules are applied consistently and history is visible to authorized users.	Planned
S4-NOT-01	ST25-ST26	Integrate notifications and message-center updates for important events.	Duy Anh	Khánh	6h	Relevant status changes create the documented notification.	Planned
S4-AI-01	AI-01	Build a labeled maintenance-ticket evaluation set and measure category, priority, and routing performance.	Đạt	Khánh	6h	Evaluation results and known failure cases are documented.	Planned

S4-AI-02	AI-01	Integrate Smart Ticketing predictions into the manager maintenance queue with confidence display and manual override.	Duy Anh	Đạt	8h	Manager can accept or correct AI output, and the correction is auditable.	Planned
S4-AI-03	AI-02	Collect and approve rulebooks, FAQs, and procedures for the RAG knowledge base.	Khánh	Đạt	5h	Every indexed source is approved, versioned, and traceable.	Planned
S4-AI-04	AI-02	Prototype retrieval, grounded response, source citation, and fallback behavior for the virtual assistant.	Duy Anh	Kiên	8h	Test questions receive grounded answers or a safe escalation response.	Planned
S4-INT-01	All operational groups	Integrate finance, residence, conduct, maintenance, and notification workflows.	Duy Anh	Đạt	8h	Regression smoke tests pass across the integrated modules.	Planned
S4-TEST-01	All operational groups	Run role, API, database, and regression tests.	Đạt	Duy Anh	8h	No unresolved critical defect remains before Build 4.	Planned

5.6 Sprint 5 Backlog – Release and Demonstration

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Task ID	Release activity	Owner	Reviewer	Estimate	Acceptance criteria	Status
S5-INT-01	Complete final frontend/backend integration and resolve high-priority defects.	Duy Anh	Đạt	10h	All accepted critical workflows pass the regression suite.	Planned
S5-SEC-01	Run authentication, authorization, input-validation, and secret-management checks.	Đạt	Duy Anh	8h	No critical unauthorized-access path remains open.	Planned
S5-PERF-01	Run basic performance checks for key APIs and dashboards.	Khánh	Duy Anh	5h	Results are recorded against the documented targets.	Planned
S5-UAT-01	Conduct user acceptance testing with representative scenarios.	Đạt	Kiên	8h	UAT scenarios are accepted or have documented limitations.	Planned
S5-DEP-01	Rehearse deployment and application-environment recovery.	Duy Anh	Khánh	5h	The application can be deployed using documented steps.	Planned

S5-DOC-01	Complete user guide, release notes, final reports, and PDF conversion.	Đạt	Duy Anh	8h	All documents are readable, consistent, and linked from README.	Planned
S5-DEMO-01	Record the narrated PA5 demonstration video.	Kiên	Đạt	5h	Video demonstrates the selected functional group and alternative flows.	Planned
S5-PKG-01	Package Markdown, PDF, source code, evidence, and Git log into the required ZIP.	Đạt	Duy Anh	4h	ZIP follows PA5 - Group[GroupId] structure and excludes generated folders.	Planned

6. Build and Test Plan

Performed by: Đào Duy Anh | *Reviewed by:* Trần Hoàng Quốc Khánh | *Edited by:* Trần Huỳnh Mạnh Đạt

The team will create five formal integration builds. Local and pull-request builds may occur more frequently but do not replace the formal build records.

Build	Target date	Sprint	Scope	Testing purpose
Build 1 — Concept Prototype	06 June 2026	Sprint 1	Initial repositories and representative interface/API foundation	Confirm technical feasibility and repository setup
Build 2 — PA2 Baseline	11 July 2026	Sprint 2	Authentication, roles, current modules, overdue notifications, and room-transfer smoke path	Create a stable documented baseline
Build 3 — PA3 Maintenance Increment	25 July 2026	Sprint 3	Maintenance requests, assignment, status updates, image evidence, Admin/Student UI improvements, and authorization regression	End-to-end maintenance workflow, role authorization, and integration testing
Build 4 — Operational Beta	08 August 2026	Sprint 4	Invoices, maintenance, absences, violations, notifications, and dashboards	Broad regression and stakeholder review
Build 5 — Final Demo Release	22 August 2026	Sprint 5	All accepted features, fixes, deployment, guides, and release notes	PA5 acceptance and complete feature demonstration

6.1 Build Record Checklist

Every formal build records the following evidence:

Evidence item	Purpose
Frontend commit	Identifies the exact frontend source state.
Backend commit	Identifies the exact backend source state.
Build identifier or tag	Gives the build a stable reference for review.
Environment configuration version	Records the runtime and deployment settings.

Database and test-data version	Keeps tests reproducible.
Test result summary	Shows what passed, failed, or needs retesting.
Open defects and known limitations	Makes unresolved risk visible.
Demonstration instructions	Allows reviewers to repeat the demo path.

6.2 Quality Gate Summary

Gate	Minimum evidence
Integration	Frontend, backend, database, and storage paths run together for the selected workflows.
Authorization	Student, Admin, Dormitory Manager, and Maintenance Staff access rules are checked.
Validation	Required fields, invalid inputs, file evidence, and status transitions are tested.
Regression	Previously completed authentication, room, profile, and notification paths remain usable.
Documentation	SDP, Vision, use-case specifications, reports, and evidence package are updated.

7. Trello, Repository, and Configuration Rules

Performed by: Đào Duy Anh | *Reviewed by:* Trần Huỳnh Mạnh Đạt | *Edited by:* Trần Huỳnh Mạnh Đạt

- Trello is the source of truth for Product Backlog order, Sprint Backlog membership, task keys, owners, reviewers, estimates, due dates, and status. [Trello board](#).
- Every coding, documentation, training, testing, and report task must exist in Trello.
- A pull request or documented peer review is required before merge.
- Frontend and backend changes for the same story should reference the same Trello item.
- API changes must update request/response examples.
- Secrets and real credentials must not be committed.
- Formal builds should use matching identifiers in both repositories.
- A capability is described as implemented only when code and validation evidence exist.

8. Monitoring and Reporting

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

Measure or report	Target / frequency
Sprint Goal progress	Reviewed at each Scrum meeting
Sprint Backlog completion	Target at least 80%; incomplete work returns to the Product Backlog
Independent review coverage	100% of tasks have a reviewer different from the owner
Frontend/backend integration	At least twice per sprint
Formal build completion	Five of five planned builds

Critical acceptance tests before PA5	100% pass
Open critical defects before Build 5	Zero
Weekly Report	Every sprint according to course instructions
AI Usage Report	Every sprint
Risk review	Weekly and at Sprint Review
Product Backlog refinement	At least once per week

9. PA3 Submission Checklist

Performed by: Trần Huỳnh Mạnh Đạt | *Reviewed by:* Đào Duy Anh | *Edited by:* Trần Huỳnh Mạnh Đạt

- ✓ Vision Document is in English Markdown.
- ✓ Software Development Plan is in English Markdown.
- ✓ Both documents have PDF versions.
- ✓ The Vision contains two valid Mermaid workflows.
- ✓ Every section includes performer, reviewer, and editor information.
- ✓ The project plan contains exactly five PA-aligned sprints.
- ✓ The current sprint contains detailed tasks, one owner, a different reviewer, and due dates.
- ✓ The build plan contains five formal builds.
- ✓ Trello dates and assignments match this document.
- ✓ Spec Kit files are inside `/src`.
- ✓ Training evidence exists for all members.
- ✓ AI Usage Report is complete.
- ✓ Weekly Report contains Sprint Planning, Scrum meetings, Sprint Review, and Trello screenshots.
- ✓ Git log evidence is included.
- ✓ The final folder is named `PA3-Group[GroupId]`.
- ✓ Markdown, PDF, reports, screenshots, and evidence are included in the ZIP.

End of Software Development Plan — Version 1.1